ISA 401: Business Intelligence & Data Visualization

12: Structured Text Extraction with LLMs

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Automated Scheduler for Office Hours

Fall 2026

Before We Start

1 Join today's Zoom

Use the Zoom link in the **Week 0 Module** on Canvas. **Mute** your mic and **turn off** your video.
Log in from the **lab's computer** so you can **share your screen** if you need help with an error.

2 Open the project, start the markdown

GitHub Desktop → `business_intelligence` → **Fetch origin**, then open `business_intelligence.Rproj`.
New R Markdown, title *Class 12*, delete everything below the setup chunk, save it as **markdowns/class12.Rmd**.

3 Check two things in the Console

`packageVersion("ellmer")` prints a version number. If it gives an error, run `install.packages("ellmer")`.
`nchar(Sys.getenv("OPENAI_API_KEY"))` prints a number above 0.
Both were set up for the app in Class 07.

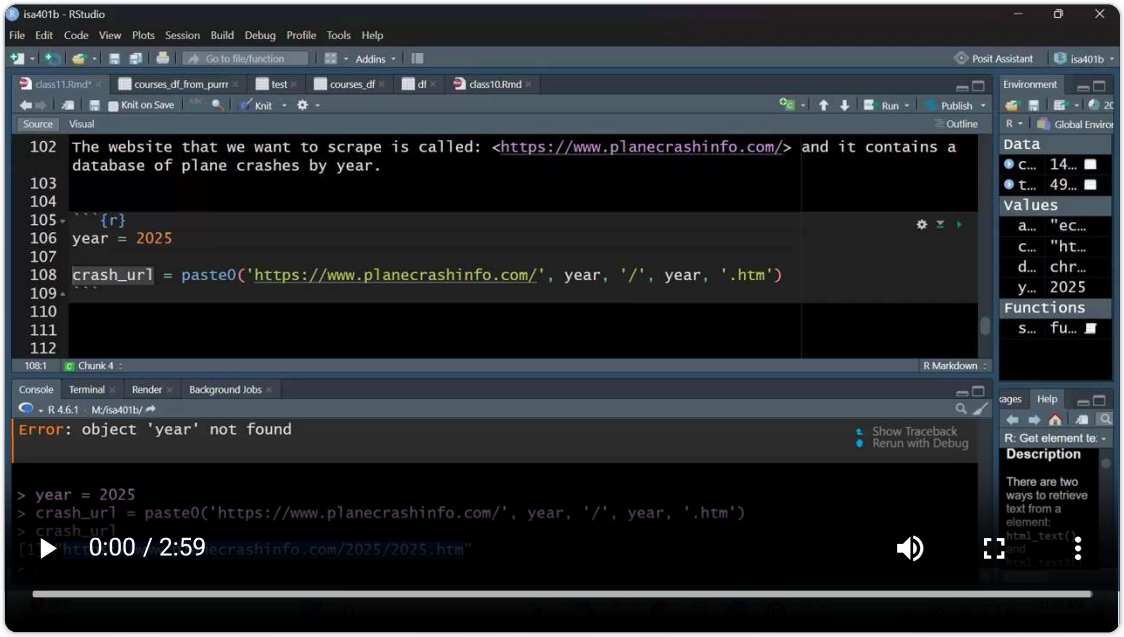
4 Today's plan

8 min. Finish the planecrashinfo example from Class 11.
20 min. Practice scraping many pages: Salvage Run.
45 min. Structured text extraction with an LLM.
Exam 01 is Monday, Oct 12. The practice exam is on the last slide.

Done early? **Help the person next to you.** Stuck? **Share your screen on Zoom** and we fix it together.

Part 1: Finishing planecrashinfo

Where We Stopped on Wednesday



Class 11 recording (Sep 30, 2026), 2 min 59 s, in three pieces: the error, the fix, and the second encoding (from 2:32).

What happened

1. `rvest::read_html(crash_url)` ran, but printing the page stopped with `invalid multibyte string`.
2. We pasted the code and the error into ChatISA. It suggested naming the encoding.
3. With `encoding = "ISO-8859-1"` the page printed and `html_table()` returned the table.
4. ChatISA's second suggestion, `encoding = "windows-1252"`, also ran without the error.

Why it happened

The page's `<head>` says `text/html; charset=windows-1252`. R expected UTF-8, so characters outside plain English letters could not be read until we named an encoding. Both suggestions worked.

The Table Still Needs Cleaning

Question. These are four of the 14 rows that `rvest::html_table()` returned for 2025. What has to change before we can analyze them?

`crash_raw (four rows)`

Date	Location / Operator	Aircraft Type / Registration	Fatalities
29 Jan 2025	Washington DCAmerican Eagle / US Army	Canadair CRJ-701ER /UH-60 helicopterN709PS /?	67/67(0)
17 Feb 2025	Toronto, CanadaDelta Connection, Edeavor Air	Bombardier CRJ-900LRN932XJ	0/80(0)
12 Jun 2025	Ahmedabad, IndiaAir India	Boeing 787-8 DreamlinerVT-ANB	241/242(19)
04 Nov 2025	Louisville, KentuckyUnited Parcel Service	McDonnell Douglas MD11FN259UP	3/3(12)

What needs cleaning

- Edit me to include any notes/discussion

Source: planecrashinfo.com/2025/2025.htm, read Oct 05, 2026. We clean this table on Wednesday.

Part 2: Scraping Many Pages

Practice: Salvage Run

15:00

Open the crew console for your section

Section A: isa401-salvage.vercel.app/?section=A

Section B: isa401-salvage.vercel.app/?section=B

Work in pairs, in `class12.Rmd`. The page you scrape is the Ceres Salvage Registry: 400 ships, 50 on each page. Enter each answer in the console to move your ship on the board.

Pause between requests. The registry's robots.txt asks for it: put `Sys.sleep(1)` inside your loop.

Three stages, one skill each

Stage 1. A CSS selector that returns one value for each of the 50 ships on the first page.

Stage 2. How the address changes when you click Next.

Stage 3. A `for` loop over every page, then one number computed from all 400 ships.

Finished early? Rewrite stage 3 as a function and `purrr::map()`.

- Edit me to include any notes/discussion

Part 3: Structured Text Extraction

Three Ways We Get Data

Classes 08 and 09: an API

structured

The owner sends JSON with named fields. We ask for a field by its name.

FRED, USAJOBS, Cincinnati open data.

Classes 10 and 11: web scraping

semi-structured

The data sits in HTML tags. A CSS selector works when each value has its own tag.

Wikipedia tables, the course catalog, planecrashinfo.

Today: text extraction

unstructured

The value is somewhere inside a sentence, an image, or a PDF. No field name and no tag points to it.

A prerequisite inside a course description. A number shown only as a bar in a chart.

Learning objectives. (1) Explain the role of LLMs in extracting structured data from text. (2) Use an LLM API from R to parse unstructured text. (3) Write a schema and field descriptions that tell the model what to return. (4) Compare LLM extraction with selectors and string patterns.

A Value Without Its Own Tag

Question. In Class 10 we scraped this course with two selectors: one for the title and one for the description. Which selector would return only the prerequisite? Only the credit hours?

One course block from the ISA catalog page, as text

ISA 401/ISA 501. Business Intelligence and Data Visualization. (3)
An introduction to the use of business intelligence and data visualization in organizations, with emphasis on how information is gathered, stored, analyzed, and used. Topics covered include business intelligence, data warehousing, data visualization, and data mining.
Prerequisite: ISA 245 or ISA 345 or CSE 385.

The columns we want

column	type
code	text
name	text
credit_hours	number
course_desc	text
prereqs	text

No selector does. The code, the name, and the credit hours share one tag, and the prerequisite is a sentence inside the description. The usual tool is a string pattern (a regular expression) written for this exact layout. A second option is to give the text to a language model and tell it which columns we want.

Text In, One Row Out

1 Input: the text of one course, stored as `course_text`

ISA 401/ISA 501 . Business Intelligence and Data Visualization . (3)
An introduction to the use of business intelligence and data visualization in organizations, with emphasis on how information is gathered, stored, analyzed, and used. Topics covered include business intelligence, data warehousing, data visualization, and data mining.
Prerequisite: ISA 245 or ISA 345 or CSE 385 .

1. Input. One course block, exactly as `rvest::html_text2()` returned it. Nothing has been sent yet.

Back | Step | Play | Reset | 0.5x | 1x | 2x

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ellmer Types and R Column Types

In the schema	What comes back in R	Example from the catalog
`ellmer::type_string()`	character	`"ISA 401"`
`ellmer::type_integer()`	integer	`3`
`ellmer::type_number()`	double	`1.5`
`ellmer::type_boolean()`	logical	`TRUE` if the course is cross-listed
`ellmer::type_enum(c("undergraduate", "graduate"))`	one of the listed values	`"graduate"`
`ellmer::type_array(...)`	several values of one type	every course named in the prerequisites
`ellmer::type_object(...)`	named fields: one row	one course

Two arguments on every type. The first is the **description**: one sentence telling the model what belongs in the field. `required = FALSE` lets the model leave the field empty when the text does not contain it; without it the model must return something.

Source: ellmer reference, Type specifications.

The ellmer Structured Data Article

ellmer 0.5.0 Get started Reference Articles News Search for

Structured data

Source: vignettes/structured-data.Rmd



When using an LLM to extract data from text or images, you can ask the chatbot to format it in JSON or any other format that you like. This works well most of the time, but there's no guarantee that you'll get the exact format you want. In particular, if you're trying to get JSON, you'll find that it's typically surrounded in ```json, and you'll occasionally get text that isn't valid JSON. To avoid these problems, you can use a recent LLM feature: **structured data** (aka structured output). With structured data, you supply the type specification that defines the object structure you want and the LLM ensures that's what you'll get back.

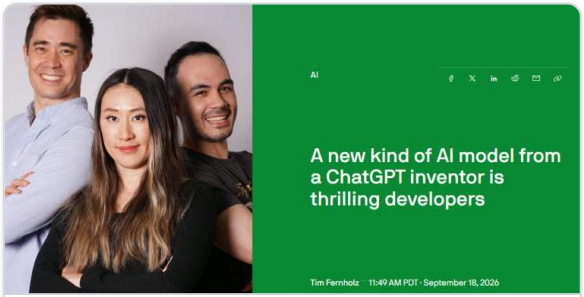
ON THIS PAGE

- Structured data basics
- Data types
- Examples
- Tools and structured data
- Token usage

In RStudio: `vignette(topic = "structured-data", package = "ellmer")`. Online: ellmer.tidyverse.org/articles/structured-data.html.

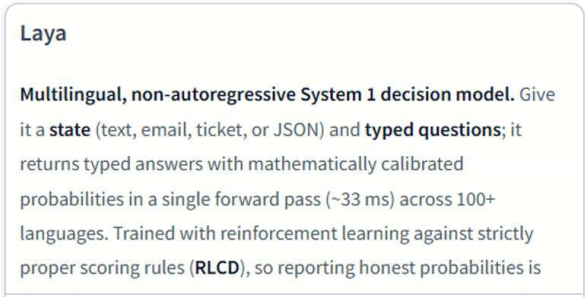
Related News: Decision Models

Three recent releases use the same idea, a schema that limits what a model may return, for one narrow job: answering questions that have a fixed set of answers.



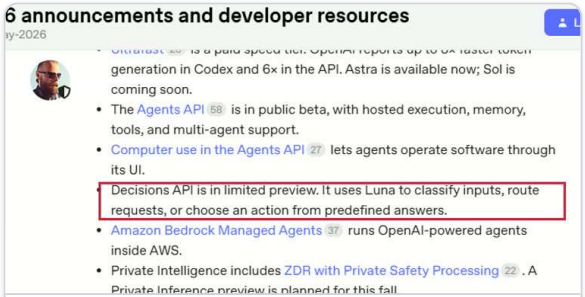
Jev, from TypeSafe AI. Early access. TypeSafe describes it as built "to make fast, structured decisions that software can use directly."

TechCrunch, Sep 18, 2026 · TypeSafe announcement · docs



Laya, from Convai Innovations. Open weights under Apache 2.0. Takes text and typed questions and returns typed answers with probabilities.

Hugging Face model card, posted Sep 18, 2026

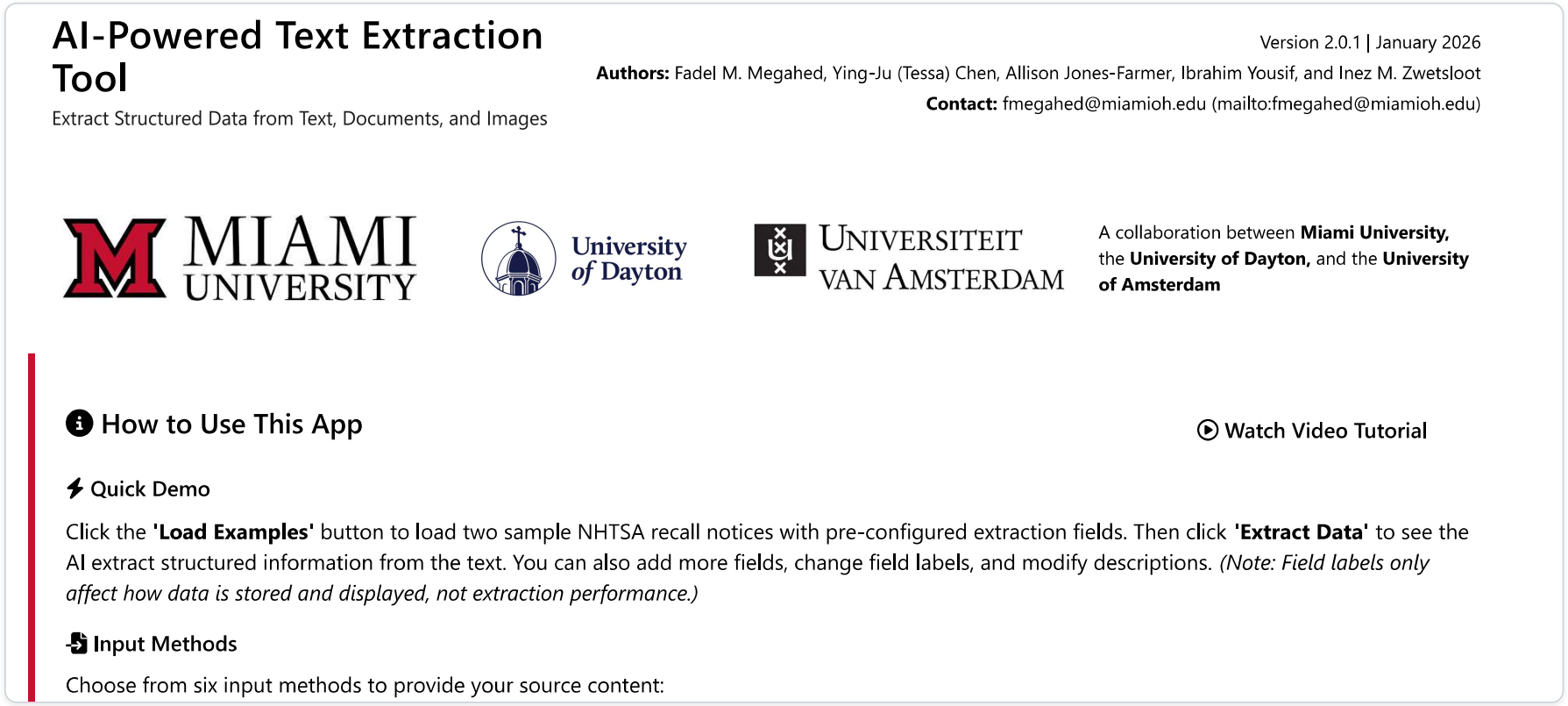


Decisions API, from OpenAI. In limited preview. It classifies inputs, routes requests, or chooses an action from answers the developer defines.

OpenAI Developer Community, Sep 29, 2026 · DevDay recap

	Structured extraction (today)	Decision models
Question	"What is the prerequisite?"	"Is this a graduate course: yes or no?"
Answer	Any text or number found in the source	One option from a list we wrote, or a score
Also returned	Nothing else	A probability for the answer

An App for Trying Schemas



Open it in its own tab: huggingface.co/spaces/fmegahed/structured_text_extraction. The app is written in R with Shiny and ellmer.

Demo 1: The ISA Catalog with an LLM

Question | Result | Cost | Checks | Discussion

Question. Build the same data frame of ISA courses as in Class 10, with three more columns: the course code, the credit hours as a number, and the prerequisites on their own.

Step 1: one course

Open the vignette, copy its first example, and make it work on the text of ISA 125 with `chat$chat_structured()`.

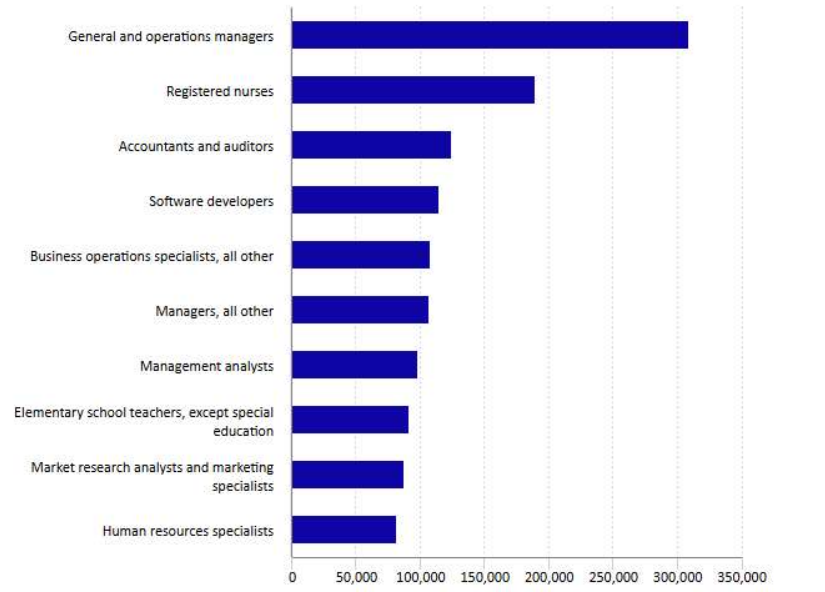
Step 2: all 50 courses

One selector, `div.courseblock`, returns each course as a single block of text. Send every block with the same schema using `ellmer::parallel_chat_structured()`.

Page: bulletin.miamioh.edu/courses-instruction/isa/. We write the code together. Type it in `class12.Rmd`.

Demo 2: A Chart Image to a Data Frame

Question | Result | Discussion



Question. This chart is a picture: there is no table and no tag to select. Build a data frame with one row per occupation and its projected openings each year.

Send the image with `ellmer::content_image_file()`. For many rows with the same columns, the schema is `ellmer::type_array()` around `ellmer::type_object()`.

Download **bls_bachelors_openings.png** from Canvas and save it in your `data` folder.

Chart 4 in Education level and projected openings, 2024-34, U.S. Bureau of Labor Statistics, Career Outlook, September 2025.

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Choosing a Method

The data is...	Use	Same result every run?	Cost	What to check
offered through an API	`httr2` (Classes 08, 09)	Yes	Free or a key	Status code, number of rows
in HTML, each value in its own tag	`rvest` and a CSS selector (Classes 10, 11)	Yes	Free	Element counts match the page
in text that follows a fixed pattern	String functions and regular expressions	Yes	Free	Rows where the pattern did not match
in free text, an image, or a PDF	An LLM with a schema (today)	No	Paid per token	A sample of rows against the source

From today's catalog run. 50 courses cost $0.048 and took 30 seconds. 14 of 47 descriptions contain wording that is not on the page, and a second run changed the prerequisites text in 12 rows. Use the model for the columns that need it, and a selector or string code for the rest.

Recap

Summary of Main Points

By now, you should be able to do the following:

- **Explain** the role of LLMs in extraction: they return values that sit inside free text or an image, where no field name or tag points to them.
- **Use** an LLM API from R: `ellmer::chat_openai()`, then `chat$chat_structured()` for one text and `ellmer::parallel_chat_structured()` for many.
- **Write** a schema: `ellmer::type_object()` with one typed field per column, a description for each field, and `required = FALSE` for values that may be absent.
- **Compare** methods: selectors and string patterns are free and give the same result every run; an LLM handles free text and images, costs money, and needs its output checked against the source.

Next class (Wed, Oct 7): Tidy, technically correct, and consistent data.

🎯 Exam 01 and Assignments 🎯

Exam 01: Monday, Oct 12

Data acquisition and transformation (Phase I on the course schedule).

Practice exam: the **Fall 2025 Exam 01** is open on Canvas. Work through it with your notes and R open, the way you will take the exam.

Before you leave: knit, then commit and push `markdowns/class12.Rmd` and `class12.html`. If it is not pushed, it is not done.

Assignment 07: FSB Directory

One question, 10 points. Scrape the faculty and staff directory pages of the seven Farmer School departments into a single data frame: department, name, position, and website.

Instructions and the due date are on the **Canvas Assignments page**.

Reading, if you want more: the ellmer articles on structured data and prompt design.